Jay Kim

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TECHNICAL SKILLS

Languages	Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks	Django REST Framework, FastAPI, Flask, React, Next.js, Node.js, Express.js, Testing (JUnit, PyTest)
Databases	PostgreSQL, MySQL, Oracle Database, MongoDB, Redis, AWS DynamoDB
Cloud/DevOps	AWS EC2, AWS S3, Microsoft Azure, Docker, Kubernetes, GitHub Actions, CI/CD
Data/ML	Apache Spark, Databricks, Snowflake, Azure Data Explorer, Pandas, PyTorch, Scikit-Learn
Tools	Git, GitHub, Jira, Postman, Splunk, Tableau, Alteryx

EXPERIENCE

Qcells — **San Francisco, CA**
Software Engineer — June 2026 – Present

- Developed FastAPI microservices exposing real time battery, inverter, and grid telemetry via REST APIs, giving downstream engineering teams standardized, equipment-level access to operational data
- Built an internal Python package automating the full API delivery lifecycle for read endpoints including a universal testing suite applied across all services, halving development time for new telemetry endpoints
- Engineered data pipelines that processed time-series data from batteries and inverters into Azure Data Explorer, supporting real-time alerting on battery state of charge and grid dispatch decisions
- Automated build, test, and deployment workflows with GitHub Actions and implemented application monitoring in Azure, reducing manual deployment steps and improving release observability

Capital One — **McLean, VA**
Python Developer & Data Analyst | Data Infrastructure, Analysis, and Engineering — August 2021 – August 2023

- Built a full-stack application using React and Python/Django REST Framework that surfaced KPIs and statistical stability metrics across 20+ bank models, replacing recurring ad hoc analyst requests with standardized automated reporting
- Engineered Apache Spark data pipelines that cleaned and transformed 100M+ rows of transaction data, reducing model execution time by 25% through targeted performance optimizations
- Developed internal Python tools that automated transaction pattern analysis for digital asset monitoring workflows, accelerating AML alert reviews by 40%
- Integrated the Holistic Risk Scoring Model into AML data pipelines and engineered predictive features from 3+ years of historical investigation outcomes, improving fraud alert escalation precision by 10%

Franklin Templeton (Western Asset Management) — **Pasadena, CA**
Python Developer Intern | Quantitative Investment Operations — June 2020 – August 2020

- Developed a portfolio data management system and dashboard by integrating internal REST APIs with Snowflake to assess and track portfolio data for 200+ clients, increasing operational efficiency by 90%
- Migrated Alteryx portfolio tracking workflows to reusable Python libraries, enabling identification of underperforming assets and allocation analysis across fixed-income strategies
- Created a portfolio back-testing engine using Python and Alteryx to measure projected returns across 10+ fixed-income derivatives and ETF strategies using historical and benchmark return data
- Streamlined client cash data management by implementing an automated data ingestion framework for uploading and downloading client data; projected to save 300 work hours annually

EDUCATION

Georgia Institute of Technology — **Atlanta, GA**
Master of Science, Computer Science (Specialization in Artificial Intelligence)
Specialization GPA: 4.0/4.0 | **Cumulative GPA:** 3.5/4.0
Relevant Coursework: Advanced Internet Computing Systems and Application Development, Software Development Process, Machine Learning, Knowledge-Based AI, Machine Learning for Trading, Security Incident Response, Health Informatics

University of Southern California — **Los Angeles, CA**
Bachelor of Science, Business Administration (Concentration in Data Science)